



QA TESTING & QUALITY ASSURANCE DASHBOARD PROJECT

A Power BI Portfolio Project — Star Schema Design, Data Modeling &
Interactive Analytics



SEPTEMBER 24, 2026

RANJANA MITRA

Contents

Project Overview	2
The Data Model.....	2
Data Quality: Found, Diagnosed, and Fixed	3
Technical Highlights in the DAX Layer	3
Dashboard Structure — 7 Pages	4
Design Decisions Worth Noting	4
Tools Used	5
Appendix: DAX Measures.....	5
Execution Metrics.....	5
Defect Metrics.....	5
Severity-Priority Mismatch Count =	6
Cross-Fact & Cumulative (Advanced).....	6
Reflection	7
Snapshot of the Project.....	8

Project Overview

This project is an end-to-end Power BI dashboard built for the IT Testing & Quality Assurance domain — covering test execution tracking, defect management, SLA compliance, release readiness, and team workload distribution across five simulated software projects (banking, e-commerce, HR, and insurance applications).

The goal wasn't just to produce charts — it was to build a **correctly modelled, production-style analytical system**: a proper star schema, verified DAX logic, and a multi-page report designed around how a QA lead would actually use it day to day.

Background: 18 years in IT, 3 years in data analysis, prior experience building dashboards in Excel with Power Query. This project was my first hands-on Power BI build, taken from raw data model through to a finished 7-page report.

The Data Model

Rather than working with pre-cleaned, flat data, this project was built on a proper **star schema**: 9 dimension tables and 2 fact tables, connected by 15 relationships (14 active, 1 deliberately inactive).

Dimensions: tbl_Dim_Date, tbl_Dim_Project, tbl_Dim_Application, tbl_Dim_Tester, tbl_Dim_Environment, tbl_Dim_Test Type, tbl_Dim_Severity, tbl_Dim_Priority, tbl_Dim_Test Case

Facts: tbl_Fact_Test Execution (6,000 rows), tbl_Fact_Defects (1,388 rows)

A few modelling decisions worth calling out specifically:

- **Severity vs. Priority were modelled as two separate, only loosely correlated dimensions** — reflecting a real QA distinction: severity is technical impact, priority is business urgency. A Critical-severity defect isn't automatically P1; a Medium-severity defect can be escalated to P1 if it blocks a release. This let the dashboard support a genuine Severity × Priority matrix rather than collapsing the two into one axis.
- **A fact-to-fact bridge relationship** (Fact_TestExecution ↔ Fact_Defects on ExecutionKey) was built and deliberately kept **inactive**, activated only inside specific DAX measures via USERELATIONSHIP(). Keeping it always-on would have created ambiguous multi-path filtering against every shared dimension (Date, Project, Application, Tester, Test Case) — Power BI's own relationship validation confirmed this directly when tested.

- **Dim_TestCase deliberately avoided snowflaking** — even though it holds ApplicationKey and TestTypeKey as descriptive foreign keys, those were kept unmodeled (no relationship drawn) to avoid redundant join paths back to the fact tables. Application and Test Type filtering flows through the fact table directly, not through the test case dimension.

Data Quality: Found, Diagnosed, and Fixed

Fixing a hidden data quality issue

While auditing the dataset, I found a flaw in how SLA breaches were being flagged. The original logic only checked whether a defect had missed its deadline after it was already fixed — so any defect that was still open, no matter how overdue, was automatically marked as "not breached." In other words, the defects most at risk — the ones sitting unresolved past their deadline right now — were the ones the system was least able to catch.

I fixed this in Power Query by rebuilding the check to work in real time: for defects still open, it now calculates how many hours have passed since they were raised and compares that against the SLA target for their severity level. A defect that's overdue is flagged as breached immediately, whether or not it's ever been fixed.

I kept the original field (renamed SLABreached_Legacy) next to the corrected one, so the before-and-after difference is easy to see side by side.

Technical Highlights in the DAX Layer

- **20+ measures**, including standard rate/count calculations (Pass Rate %, Test Coverage %, Defect Density, Automation Rate %) and more advanced patterns:
- **A true point-in-time cumulative measure** (Open Defects (Cumulative) / Closed Defects (Cumulative)) powering a genuine defect burndown chart — using VAR, REMOVEFILTERS, and RELATED to reconstruct "how many defects were open as of this specific month," not just "defects raised this month." This is materially harder than a simple filtered count, and was verified row-by-row against the source data before trusting it.
- **A cross-fact-table measure** (Defects per Build Trend) using USERELATIONSHIP() to bridge Fact_TestExecution and Fact_Defects on demand — including debugging a real cardinality/cross-filter-direction conflict that Power BI's relationship engine flagged directly (a 1:1 relationship cannot be set to single-direction filtering; resolving it required reframing the relationship as many-to-one).

Dashboard Structure — 7 Pages

Page	Purpose
QA Executive Summary	Top-line KPIs, pass rate trend, execution status breakdown
Test Coverage	Pass rate by application × test type, execution workload by tester
QA Performance Bottleneck	Slowest test cases, duration by module, manual vs. automated execution time
Defect Overview	Defects by severity, root cause, application, and the severity × priority matrix
Defect Ageing & SLA	SLA breach tracking (using the corrected logic), resolution time by severity, an ageing table of currently-open defects
Release & Sprint Readiness	Pass rate and defects trended by sprint, the defect burndown chart
Team Workload & Coverage	Execution and defect-reporting distribution across the QA team

Every KPI and chart was cross-verified against the underlying source data — not just visually checked, but numerically reconciled (e.g., confirming stacked bar segments sum to totals, confirming filtered subsets add up correctly against known full-dataset counts).

Design Decisions Worth Noting

- **A consistent semantic colour system** was applied throughout: green = good/resolved, red = critical/failing, amber = warning/blocked, grey = neutral. This required manually overriding Power BI's default alphabetical colour assignment on every categorical chart (Severity, Execution Status) rather than relying on automatic theme colours.
- **The Team Workload page was deliberately framed to avoid a "leaderboard" effect.** Tester-level charts are sorted alphabetically by name rather than by value, and the page is titled around workload and coverage rather than performance ranking — a

conscious choice to keep the data honest without implying a ranking that raw execution counts don't actually support.

- **A Top N filter was added to the module-duration chart** specifically so it degrades gracefully under an unfiltered "All Projects" view, rather than becoming unreadable — paired with sort-by-value ranking so the cutoff reflects genuine severity of the bottleneck, not an arbitrary count.

Tools Used

Power BI Desktop · DAX · Power Query (M) · Excel (source data staging) · custom report theme (JSON)

Appendix: DAX Measures

Execution Metrics

1. Total Executions = COUNTROWS(tbl_Fact_TestExecution)
2. Pass Count = CALCULATE(COUNTROWS(tbl_Fact_TestExecution),
tbl_Fact_TestExecution[ExecutionStatus] = "Pass")
3. Fail Count = CALCULATE(COUNTROWS(tbl_Fact_TestExecution),
tbl_Fact_TestExecution[ExecutionStatus] = "Fail")
4. Blocked Count = CALCULATE(COUNTROWS(tbl_Fact_TestExecution),
tbl_Fact_TestExecution[ExecutionStatus] = "Blocked")
5. Not Executed Count = CALCULATE(COUNTROWS(tbl_Fact_TestExecution),
tbl_Fact_TestExecution[ExecutionStatus] = "Not Executed")
6. Pass Rate % = DIVIDE([Pass Count], [Total Executions])
7. Fail Rate % = DIVIDE([Fail Count], [Total Executions])
8. Test Coverage % = DIVIDE([Total Executions] - [Not Executed Count], [Total Executions])
9. Avg Execution Duration (min) =
AVERAGE(tbl_Fact_TestExecution[ExecutionDurationMinutes])
10. Automation Rate % = DIVIDE(
CALCULATE(DISTINCTCOUNT(tbl_Fact_TestExecution[TestCaseKey]),
tbl_Dim_TestCase[ExecutionMode] = "Automated"),
DISTINCTCOUNT(tbl_Fact_TestExecution[TestCaseKey])
)

Defect Metrics

1. Total Defects = COUNTROWS(tbl_Fact_Defects)

2. Open Defects = CALCULATE(COUNTROWS(tbl_Fact_Defects),
tbl_Fact_Defects[DefectStatus] = "Open")
3. Closed Defects = CALCULATE(COUNTROWS(tbl_Fact_Defects),
tbl_Fact_Defects[DefectStatus] = "Closed")
4. Critical Open Defects =
CALCULATE(
COUNTROWS(tbl_Fact_Defects),
tbl_Fact_Defects[DefectStatus] = "Open",
tbl_Dim_Severity[SeverityName] = "Critical"
)
5. P1 Open Defects =
CALCULATE(
COUNTROWS(tbl_Fact_Defects),
tbl_Fact_Defects[DefectStatus] = "Open",
tbl_Dim_Priority[PriorityName] = "P1 - Urgent"
)
6. Avg Resolution Time (hrs) = AVERAGE(tbl_Fact_Defects[ResolutionHours])
7. Defect Density (per TC) = DIVIDE([Total Defects],
DISTINCTCOUNT(tbl_Fact_Defects[TestCaseKey]))
8. Reopen Rate % = DIVIDE(CALCULATE([Total Defects], tbl_Fact_Defects[DefectStatus]
= "Reopened"), [Total Defects])
9. SLA Breach % = DIVIDE(CALCULATE([Total Defects], tbl_Fact_Defects[IsSLABreached]),
[Total Defects])

Severity-Priority Mismatch Count =

1. CALCULATE(COUNTROWS(tbl_Fact_Defects), tbl_Fact_Defects[SeverityKey] <>
tbl_Fact_Defects[PriorityKey])
2. QA Efficiency Index = DIVIDE([Pass Count], [Total Defects] + 1)

Cross-Fact & Cumulative (Advanced)

1. Defects per Build Trend =
CALCULATE(
[Total Defects],
USERRELATIONSHIP(tbl_Fact_TestExecution[ExecutionKey], tbl_Fact_Defects[ExecutionKey])
)

```

2. Open Defects (Cumulative) =
VAR CurrentDate = MAX(tbl_Dim_Date[Date])
RETURN
CALCULATE(
    COUNTROWS(tbl_Fact_Defects),
    REMOVEFILTERS(tbl_Dim_Date),
    FILTER(
        tbl_Fact_Defects,
        RELATED(tbl_Dim_Date[Date]) <= CurrentDate
        && (ISBLANK(tbl_Fact_Defects[ResolutionDate]) || tbl_Fact_Defects[ResolutionDate]
> CurrentDate)
    )
)

```

```

3. Closed Defects (Cumulative) =
VAR CurrentDate = MAX(tbl_Dim_Date[Date])
RETURN
CALCULATE(
    COUNTROWS(tbl_Fact_Defects),
    REMOVEFILTERS(tbl_Dim_Date),
    FILTER(
        tbl_Fact_Defects,
        RELATED(tbl_Dim_Date[Date]) <= CurrentDate
        && NOT(ISBLANK(tbl_Fact_Defects[ResolutionDate]))
        && tbl_Fact_Defects[ResolutionDate] <= CurrentDate
    )
)

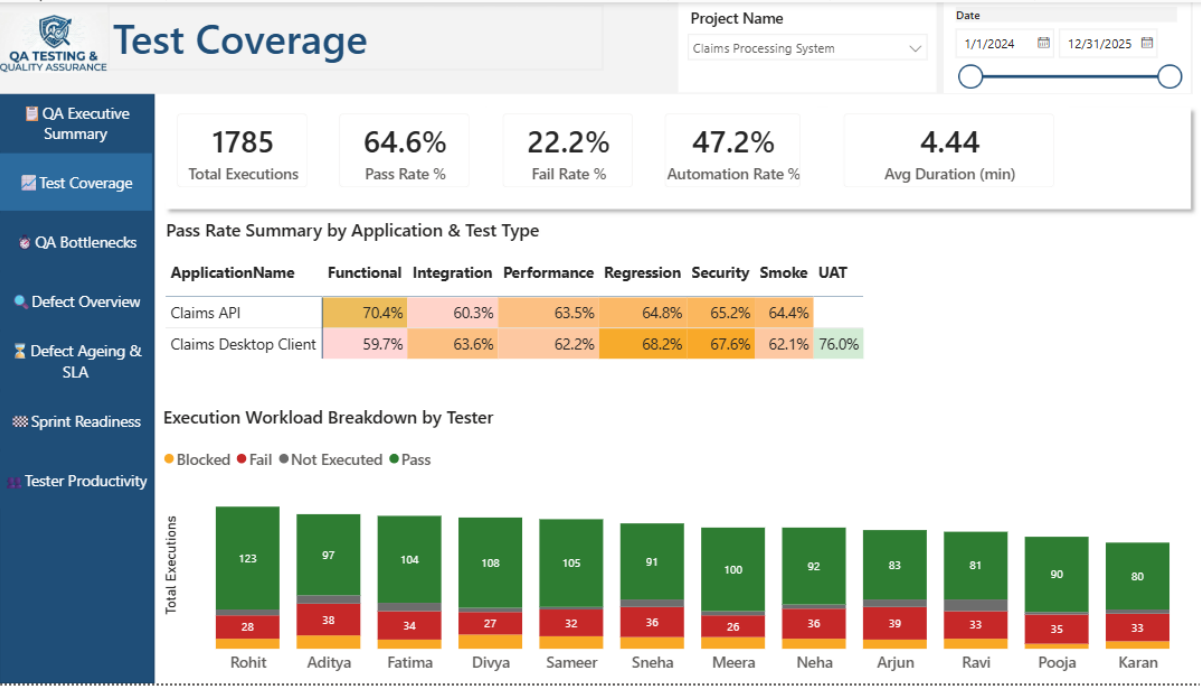
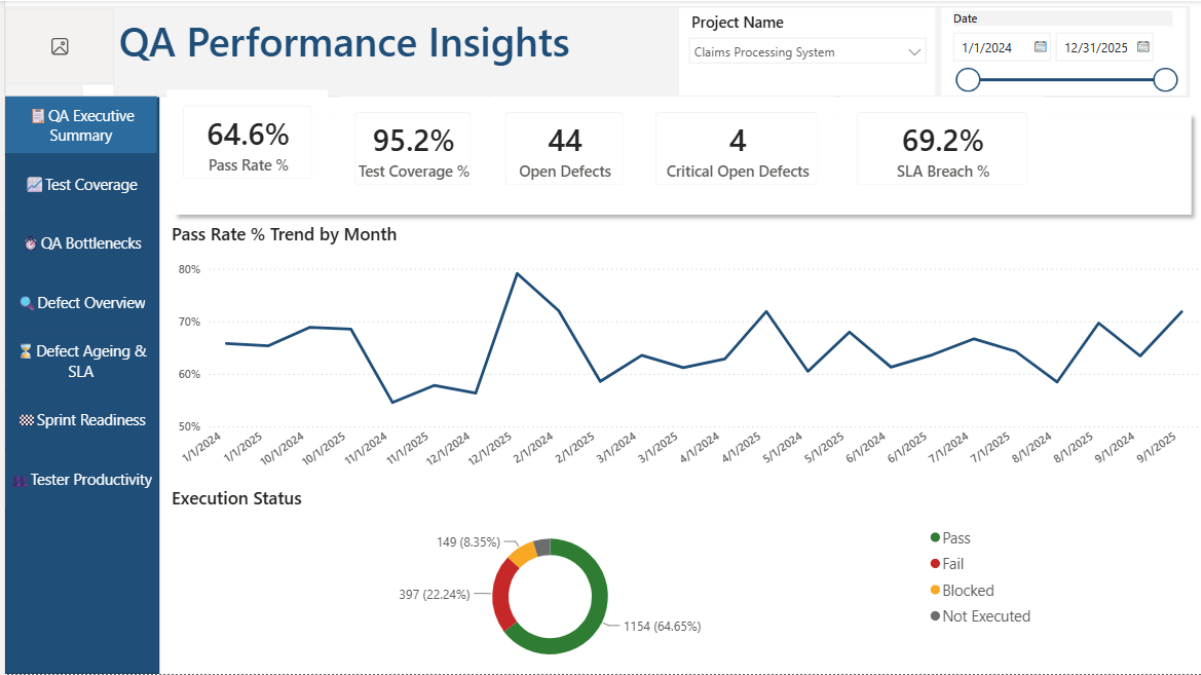
```

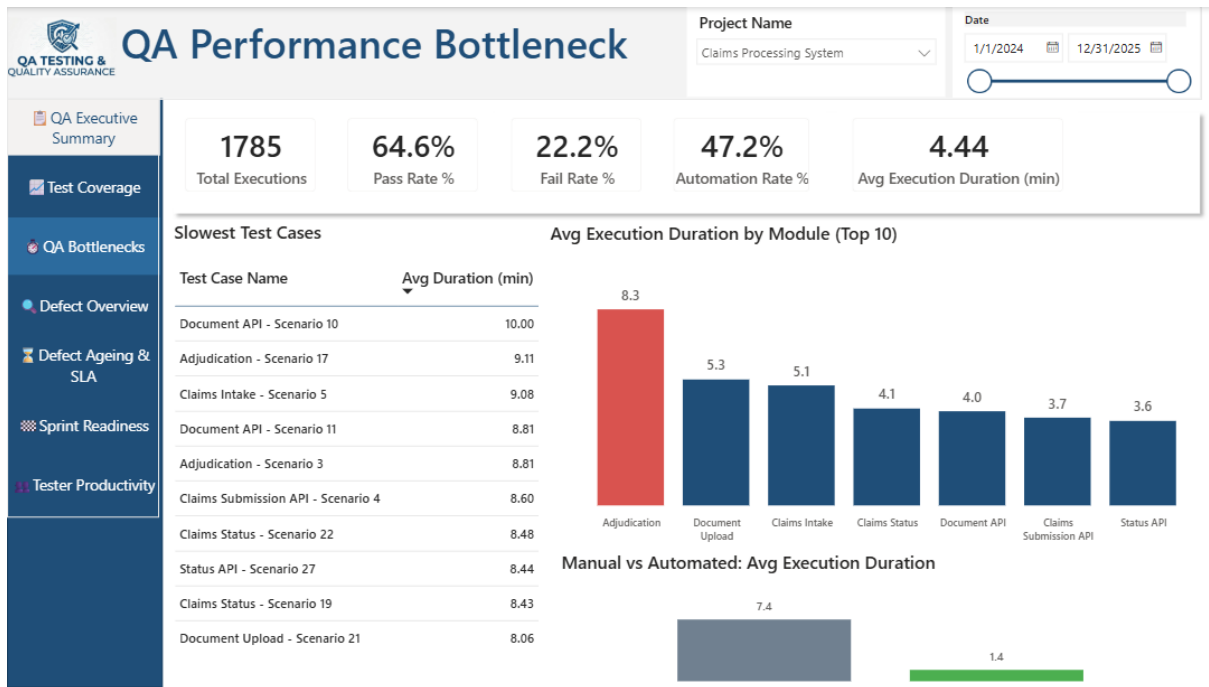
Two temporary validation measures (Orphan Check, Orphan Check Defects) were also built during development to confirm referential integrity between fact and dimension tables — used to verify data quality, not part of the report's analytical output.

Reflection

This project intentionally went beyond building charts against clean data — sourcing and auditing the dataset, correcting a real logic flaw in it, and working through several genuinely non-obvious Power BI modelling issues (ambiguous relationship paths, cardinality constraints, auto-generated date hierarchies silently overriding intended sort orders) gave this project the shape of a real analytical engagement rather than a tutorial exercise.

Snapshot of the Project





Defect Ageing & SLA

Project Name

Claims Processing System

Date

1/1/2024

12/31/2025

QA Executive Summary

Test Coverage

QA Bottlenecks

Defect Overview

Defect Ageing & SLA

Sprint Readiness

Tester Productivity

69.2%

SLA Breach %

61.89

Avg Resolution Time (hrs)

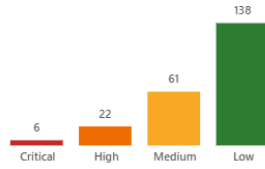
44

Open Defects

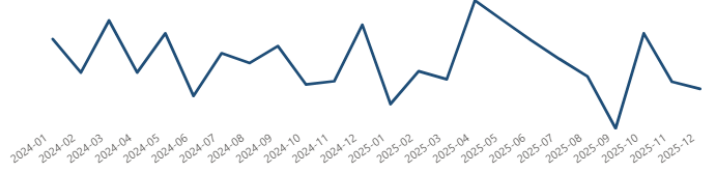
6

P1 Open Defects

Avg Resolution Time by Severity



SLA Breach % Trend



Defect Ageing Table

DefectKey	Test Case Name	Severity Name	Days Open	Priority Name	Tester Name
49	Claims Intake - Scenario 11	High	557	P1 - Urgent	Ravi Kumar
65	Status API - Scenario 13	Critical	483	P1 - Urgent	Meera Iyer
80	Claims Intake - Scenario 5	Critical	342	P1 - Urgent	Rohit Verma
192	Claims Intake - Scenario 20	Critical	952	P1 - Urgent	Pooja Desai
304	Claims Status - Scenario 8	Critical	608	P1 - Urgent	Rohit Verma
365	Document API - Scenario 11	Critical	687	P1 - Urgent	Neha Joshi
535	Claims Status - Scenario 8	Medium	932	P1 - Urgent	Rohit Verma

Release & Sprint Readiness

Project Name

Claims Processing System

Date

1/1/2024

12/31/2025

QA Executive Summary

Test Coverage

QA Bottlenecks

Defect Overview

Defect Ageing & SLA

Sprint Readiness

Tester Productivity

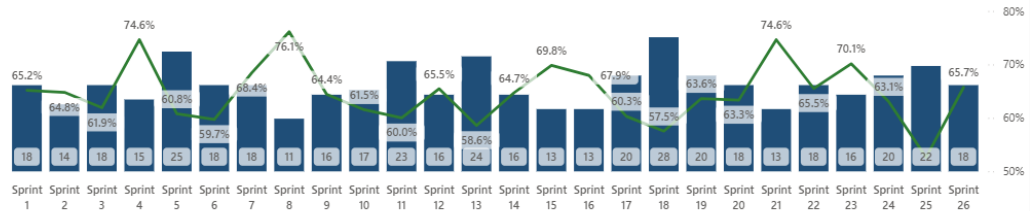
64.6%

Pass Rate %

4

Critical Open Defects

Pass Rate % and Defects trended by Sprint



Defect Burndown: Open vs Closed Over Time

● Open Defects (Cumulative) ● Closed Defects (Cumulative)

